

MOHAMMAD MAROOF

maroof.tech

+91 7418754473 | mmaroof487@gmail.com | linkedin.com/in/mohammad-marooof | github.com/mmaroof487

SUMMARY

Full Stack Engineer specializing in Next.js, React, TypeScript, Node.js, and Python, with strong expertise in secure API design, cryptography, distributed systems, and cloud-native architecture. Currently contributing to an enterprise-grade AI cybersecurity platform at SENTRIX, working across backend systems, frontend engineering, DevSecOps, and observability pipelines. Experienced in building scalable, production-ready systems with a focus on security, performance optimization, and reliability.

EXPERIENCE

SENTRIX | TALENCIAGLOBAL

Mar 2026 – Present

Intern

Chennai, India

- Engineered the Governance Layer using FastAPI and Open Policy Agent (OPA), implementing policy-based controls to mitigate prompt injection and data exfiltration attacks for AI agents.
- Designed and implemented the Telemetry Layer using OpenTelemetry, Prometheus, Grafana, Elasticsearch, and Kibana, enabling end-to-end tracing, centralized logging, and real-time monitoring.
- Developed behavioral risk scoring and audit workflows to automate policy enforcement, improve AI observability, and accelerate security incident investigation.

XENCRYPT TECHNOLOGIES

Dec 2025 – Mar 2026

Frontend Development Intern

Chennai, India

- Built and deployed the company's Next.js 16 website, integrating interactive 3D experiences while optimizing rendering, asset delivery, and performance to improve Lighthouse scores and user experience.
- Architected Encrypti-V by designing REST APIs, implementing the Node.js/TypeScript backend, engineering the PostgreSQL schema, and building the encrypted mail frontend.
- Designed scalable PostgreSQL data models and optimized queries, indexing strategies, and relational schemas to support secure messaging, file storage, and high-performance backend services.
- Implemented zero-knowledge encrypted file sharing using AES-256-GCM, ML-KEM, and chunked client-side processing for secure, memory-efficient large file transfers.

PROJECTS

Argus — Secure Intelligent Query Gateway | *Python, FastAPI, PostgreSQL, Redis, React, Docker*

- Engineered a production-ready database middleware intercepting SQL queries through a modular 6-layer pipeline spanning security, AI guardrails, caching, execution, and envelope encryption.
- Architected a zero-downtime envelope encryption subsystem (AES-256-GCM) with live key rotation and async migration, using SQL AST parsing to automatically block injection attacks and honeypot traps.
- Designed a deterministic query fingerprinting engine backed by Redis, achieving sub-5ms cache hits with intelligent table-tagged invalidation and automated downstream circuit breaking.
- Integrated Groq LLMs for natural language-to-SQL translation with strict fallback guardrails, alongside a React frontend featuring real-time observability, async audit logging, and interactive schema graphs.

Cognita — AI Engineering Intelligence Platform | *Python, FastAPI, LangGraph, React, PostgreSQL, Redis, Celery*

- Architected and built a multi-tenant SaaS platform that analyzes GitHub activity to identify engineering bottlenecks, developer burnout, and delivery risks.
- Engineered a deterministic 4-node LangGraph agent pipeline for anomaly detection, LLM-powered insight generation, and strict human-in-the-loop (HITL) remediation workflows.
- Implemented secure, async backend services featuring GitHub OAuth, AES-256-GCM encrypted credential storage, immutable audit logging, and automated integrations with Jira and Slack.

TECHNICAL SKILLS

- Languages:** Python, TypeScript, JavaScript, C++, Rust
- Frontend:** Next.js, React.js, Tailwind CSS, HTML5, CSS3
- Backend:** Node.js, Express.js, FastAPI, REST APIs, Microservices
- Databases:** PostgreSQL, MongoDB, MySQL, Redis
- DevOps & Observability:** OpenTelemetry, Prometheus, Grafana, ELK Stack, Docker, CI/CD, GitHub Actions
- Tools:** Git, GitHub, Postman, Figma

EDUCATION

SRM Institute of Science and Technology

Chennai, India

B.Tech in Computer Science and Engineering | **CGPA: 9.76**

2023 – Present